CHAPTER 18 · REFERENCE DEPTH · PARTLY NEW

Area Control & Oceanic

Once clear of the terminal area, an aircraft cruises through the airways under area control — reporting its position, holding when traffic demands, and joining, leaving or crossing routes on clearance. Out over the ocean, where there is no radar and no VHF, the same job is done by HF position reports, datalink and the procedural rules this chapter explains.

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SYLLABUS MAP

Part III (vii) Area control – units, position/level info, joining/leaving/crossing airways, holding, ATS surveillance, ADS, oceanic control

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◎ Learning objectives — by the end of this chapter you will be able to...

- Describe area control units and the structure of a position report.
- Use level-information phraseology (maintain, climb/descend, cruise).
- Use joining, leaving and crossing-airway phraseology.
- Enter and fly an en-route hold, and understand the EAT/onward-clearance system.
- Explain ATS surveillance, ADS-C/ADS-B and CPDLC in area control.

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- Describe oceanic control — the oceanic clearance, Mach-number technique and SELCAL.

18.1 Area control units

18.2 Position reporting

18.3 Level information

18.4 Joining, leaving & crossing airways

18.5 En-route holding

18.6 ATS surveillance & ADS

18.7 Oceanic control

☆ Numbers to memorise

? Question bank



En-route oceanic control relies on procedural separation, position reporting, and datalink technologies since continuous VHF radar coverage is not available.

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18.1 Area control units

IN PLAIN TERMS

The **Area Control Centre (ACC)**, with the call-sign suffix **"Control"**, provides air traffic control to aircraft **en route** in the airways and upper airspace of an FIR/UIR. A large FIR is split into **sectors**, and as you fly across it you are transferred from sector to sector.

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18.2 Position reporting

THE STANDARD POSITION REPORT

Where there is no radar, the aircraft reports its progress at compulsory reporting points. The order is: **(1) position, (2) time, (3) level, (4) next point, (5) estimate for the next point** — and the name of the point after that. ("ABC, PAPA at four two, flight level three five zero, ROMEO five five, next SIERRA.")

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TRANSCRIPT — POSITION REPORT

A/C Chennai Control, Ghostair Alfa Bravo Charlie, PAPA at four two, flight level three five zero, estimating ROMEO five five, SIERRA next.

ATC Ghostair Alfa Bravo Charlie, Chennai Control, roger, report ROMEO.

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MNEMONIC — POSITION REPORT ORDER

"Place, Time, Level, Next, Estimate (+ the one after)." Same order every time so the controller can write it straight onto the flight progress strip.

18.3 Level information

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Phrase	Meaning
"Maintain FL350"	Stay at that level
"Climb / descend to FL…"	Change to the new level
"Climb to reach FL… by [point]"	Be at the level by the point
"Cruise climb"	Climb gradually as weight reduces (oceanic)
"Report leaving / reaching FL…"	Tell ATC at the stated event
"Expedite climb/descent"	Climb/descend at best rate

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⚠ EXAM TRAP

All level instructions are **mandatory read-backs**. "Climb FL350" and "maintain FL350" are different instructions — read back exactly what was said, and do not climb or descend on a "report" or "expect" alone.

18.4 Joining, leaving & crossing airways

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THREE AIRWAY INTERACTIONS

Joining: an aircraft entering a controlled airway needs a clearance — "cleared to join airway [W] at [point], flight level …". **Leaving:** on leaving controlled airspace you are told "leaving controlled airspace, report …". **Crossing:** an aircraft crossing an airway is given a clearance to cross at a level/time keeping it separated from airway traffic.

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TRANSCRIPT — JOINING AN AIRWAY

A/C Mumbai Control, VT-ABC, request join airway WHISKEY one zero at SIERRA, flight level one five zero.

ATC VT-ABC, cleared to join airway WHISKEY one zero at SIERRA, flight level one five zero, report SIERRA.

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A/C Cleared to join WHISKEY one zero at SIERRA, flight level one five zero, will report SIERRA, VT-ABC.

18.5 En-route holding

WHY HOLD

When traffic ahead must be sequenced — for example a busy arrival airport — ATC may **hold** an aircraft in a racetrack pattern over a fix or beacon until it can be cleared onward.

HOLDING PHRASEOLOGY

"Hold at [fix], flight level …, expect onward clearance at [time]" or **"expect approach time [time] (EAT)"**. The clearance specifies the fix, the level, and the time you may expect to be released (the **Onward Clearance Time** or, for an arrival, the **Expected Approach Time, EAT**). The aircraft reads it back and flies the published hold.

TRANSCRIPT — HOLDING

ATC Ghostair Alfa Bravo Charlie, hold at PAPA, flight level one zero zero, expect approach time three five.

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A/C Hold at PAPA, flight level one zero zero, expect approach time three five, Ghostair Alfa Bravo Charlie.

18.6 ATS surveillance & ADS

ATS SURVEILLANCE — RADAR & ADS-B

In well-served continental airspace, area control uses **ATS surveillance** — primary/secondary radar and increasingly **ADS-B** (Chapter 12) — to see aircraft directly, reducing the need for position reports and allowing radar separation and vectoring.

ADS-C & CPDLC FOR REMOTE/OCEANIC AIRSPACE

Where there is no radar, **ADS-C (Automatic Dependent Surveillance — Contract)** has the aircraft automatically send position reports over datalink at agreed intervals/events, and **CPDLC (Controller–Pilot Data Link Communications)** carries clearances and messages as text. Together they let an oceanic controller "see" and instruct aircraft without voice or radar.

18.7 Oceanic control

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FIRST PRINCIPLES — CONTROL WITHOUT RADAR OR VHF

Over the ocean there is no VHF (line-of-sight) and traditionally no radar. So separation is **procedural**: aircraft are pre-cleared onto a track at a level and a **Mach number**, report their position by **HF voice or CPDLC**, and the controller keeps them apart by time and the Mach-number technique. This is why your original notes' "oceanic" was only a propagation footnote — the *control* procedure is here.

THE OCEANIC CLEARANCE & MACH-NUMBER TECHNIQUE

Before entering oceanic airspace the aircraft obtains an **oceanic clearance**: the track, the flight level, and the **assigned Mach number**. Aircraft on the same track at the same level fly the same Mach number so the time-spacing between them stays constant — the **Mach-number technique**. **SELCAL** (Chapter 22) lets the crew stop monitoring HF static until called.

TRANSCRIPT — OCEANIC CLEARANCE (OFTEN VIA HF/CPDLC)

A/C Oceanic Control, Ghostair Alfa Bravo Charlie, request oceanic clearance, estimating SIERRA at five five, requesting flight level three five zero, Mach decimal eight zero.

ATC Ghostair Alfa Bravo Charlie, cleared to destination via track SIERRA, flight level three five zero, Mach decimal eight zero, report SIERRA.

A/C Cleared via track SIERRA, flight level three five zero, Mach decimal eight zero, will report SIERRA, Ghostair Alfa Bravo Charlie.

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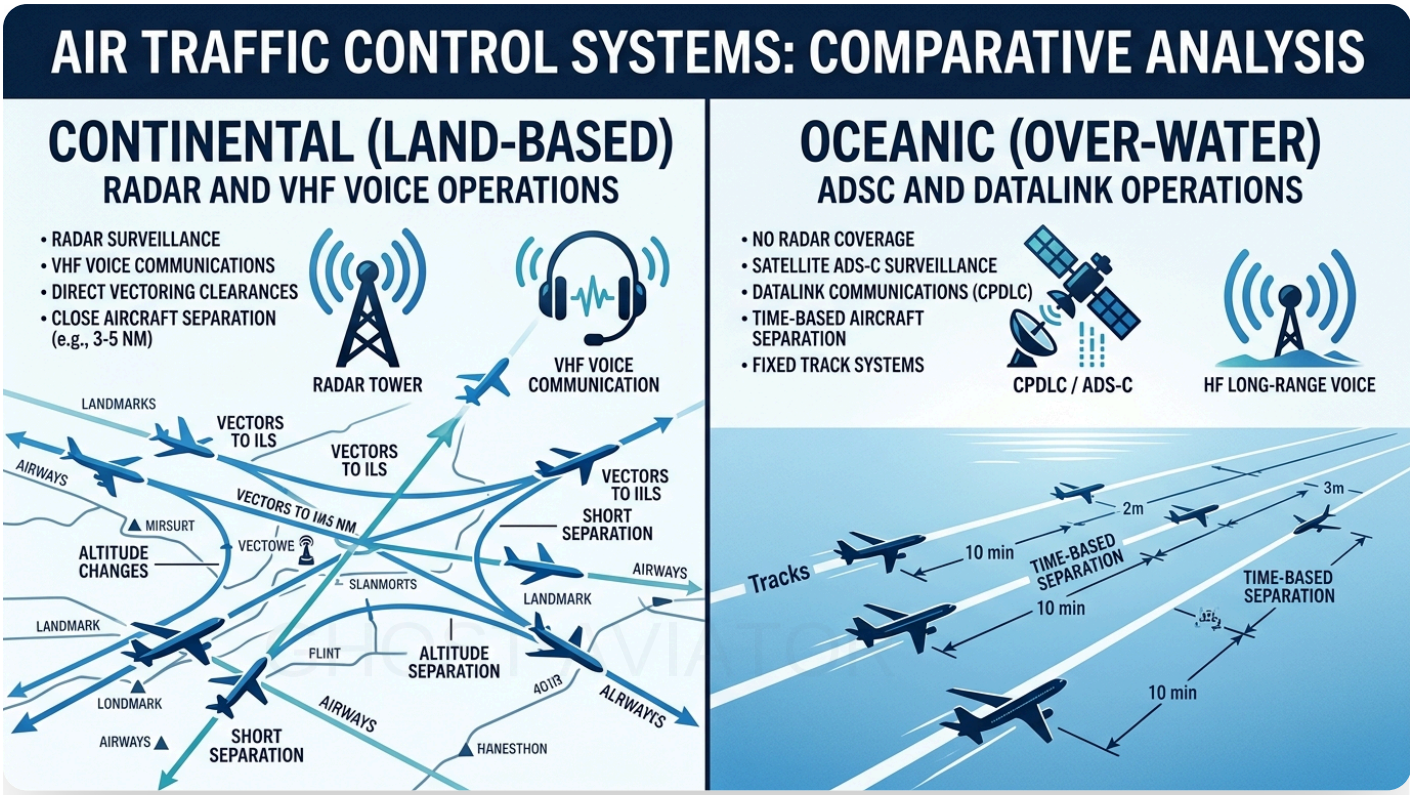


Figure 18.1 — Continental vs oceanic control: Continental airspace relies on VHF and radar vectoring. Oceanic relies on procedural tracks, CPDLC datalinks, and Mach-number spacing.

☆ Numbers to memorise

Essential Facts for Chapter 18

Fact	Value
Area Control Centre	Suffix "Control"; en-route control of the airways/upper airspace, by sector
Position report order	Position · time · level · next point · estimate (+ the one after)

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Fact	Value
Level instructions	Maintain / climb / descend — all mandatory read-backs
Airways	Join (clearance) · leave (report) · cross (level/time)
Holding	"Hold at [fix], FL…, expect approach time / onward clearance [time]"
ADS-C / CPDLC	Automatic datalink position reports / text clearances (oceanic-remote)
Oceanic	Procedural; oceanic clearance = track + level + Mach; HF/CPDLC; SELCAL

🖉 Question bank

Part A — MCQs (click an option to check)

1. The call-sign suffix for an Area Control Centre is:

Approach

Tower

Control

Information

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Answer: Control.

"Control" denotes the Area Control Centre (en-route).

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2. The correct order of a position report is:

Level, position, time, next

Position, time, level, next point, estimate

Time, next, level, position

Estimate, position, level

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Answer: Position, time, level, next point, estimate.

Position, time, level, next point and estimate (plus the point after).

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3. "Maintain flight level 350" must be:

Acknowledged "roger"

Read back

Not read back

Read back only at night

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Answer: Read back.

Level instructions are mandatory read-back items.

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4. To enter a controlled airway, an aircraft needs:

No clearance

A clearance to join

Only a squawk

A landing clearance

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Answer: A clearance to join.

Joining controlled airspace requires a clearance to join.

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5. An aircraft held over a fix is given:

A take-off clearance

A fix, level and an expected approach / onward clearance time

A SELCAL code

A runway

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Answer: A fix, level and an expected approach / onward clearance time.

Holding clearance gives the fix, level and EAT/onward-clearance time.

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6. "Expect approach time three five" tells the pilot:

To begin the approach now

The time to expect release from the hold for approach

To squawk 3500

To climb to FL350

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Answer: The time to expect release from the hold for approach.

The EAT is the time the aircraft can expect to leave the hold for its approach.

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7. ADS-C provides:

Voice communication

Automatic datalink position reports at agreed intervals/events

A glidepath

Engine data to ATC

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Answer: Automatic datalink position reports at agreed intervals/events.

ADS-C (Contract) sends automatic position reports over datalink.

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8. CPDLC carries:

Weather radar images

Controller–pilot clearances and messages as text

A glidepath signal

The phonetic alphabet

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Answer: Controller–pilot clearances and messages as text.

CPDLC is a text datalink for ATC clearances and messages.

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9. Over the ocean, separation is primarily:

By primary radar

Procedural (time/Mach), with HF/CPDLC reporting

By VHF vectoring

Not provided

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Answer: Procedural (time/Mach), with HF/CPDLC reporting.

No radar/VHF — separation is procedural, using position reports and the Mach-number technique.

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10. An oceanic clearance specifies:

Runway and squawk

Track, flight level and assigned Mach number

A heading only

A holding fix

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Answer: Track, flight level and assigned Mach number.

The oceanic clearance gives the track, level and Mach number.

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11. The Mach-number technique keeps aircraft separated by:

Vectoring

Maintaining a constant time spacing through assigned Mach numbers

Squawk codes

Radio checks

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Answer: Maintaining a constant time spacing through assigned Mach numbers.

Same Mach on the same track keeps the time interval between aircraft constant.

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12. SELCAL allows an oceanic crew to:

Vector themselves

Stop continuously monitoring HF until selectively called

Land without a clearance

Change their Mach number

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Answer: Stop continuously monitoring HF until selectively called.

SELCAL alerts a specific aircraft, so the crew need not strain at HF static.

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13. A large FIR is divided for control into:

Runways

Sectors

Aprons

Holding fixes only

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Answer: Sectors.

An ACC works the FIR in sectors, handing aircraft on between them.

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14. "Report leaving flight level 350" requires the pilot to:

Read back nothing

Tell ATC when leaving FL350

Maintain FL350

Squawk ident

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Answer: Tell ATC when leaving FL350.

A "report leaving/reaching" instruction asks for a call at that event.

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15. ATS surveillance in continental area control commonly uses:

Only HF

Radar and ADS-B

SELCAL

Light signals

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Answer: Radar and ADS-B.

Continental area control uses radar and increasingly ADS-B.

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16. An aircraft crossing an airway is cleared to cross at:

Any level it likes

A level/time that keeps it separated from airway traffic

Ground level

The transition altitude only

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Answer: A level/time that keeps it separated from airway traffic.

A crossing clearance specifies a level/time providing separation.

Part B — Oral / viva (tap to reveal model answers)

▼ ? **What does an Area Control Centre do, and what is its suffix?**

Model Answer:

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It provides air traffic control to aircraft en route in the airways and upper airspace of an FIR, worked in sectors. Its call-sign suffix is "Control".

▼ ? **State the order of a standard position report.**

Model Answer:

Position, time over it, level, the next reporting point, the estimate for that point, and the name of the point after.

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▼ ? How is an aircraft cleared to join, leave or cross an airway?

Model Answer:
Joining requires a clearance to join at a point and level; leaving controlled airspace is acknowledged with a report; crossing is cleared at a level or time that keeps the aircraft separated from airway traffic.

▼ ? What is contained in a holding clearance?

Model Answer:
The holding fix, the level, and the time to expect release — the onward clearance time or, for an arrival, the expected approach time (EAT).

▼ ? How is air traffic controlled over the ocean without radar or VHF?

Model Answer:
Procedurally: aircraft receive an oceanic clearance (track, level, Mach number), report their position by HF voice or CPDLC and via ADS-C, and are separated by time using the Mach-number technique, with SELCAL alerting the crew.

▼ ? What are ADS-C and CPDLC?

Model Answer:
ADS-C is Automatic Dependent Surveillance — Contract, where the aircraft automatically sends position reports over datalink; CPDLC is Controller–Pilot Data Link Communications, carrying clearances and messages as text. Both are used in remote/oceanic airspace.

⚡ 60-SECOND REVISION CARD

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Area Control = "Control"; en-route, worked in sectors.

Position report: **position · time · level · next · estimate**. Level instructions = mandatory read-back.

Airways: **join** (clearance) · **leave** (report) · **cross** (level/time). Hold: fix + FL + **EAT/onward time**.

Continental surveillance = **radar/ADS-B**; remote/oceanic = **ADS-C + CPDLC**.

Oceanic = procedural: **oceanic clearance (track, level, Mach)**, HF/CPDLC, **Mach-number technique**, SELCAL.

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